Project Title: Sales Performance Dashboard

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Batch/Program: B.Tech

GitHub Repository:https://github.com/ANSHUMAAN2023/Sales_Performance_Dashboard.git

1. Problem Statement

In today's competitive retail environment, businesses generate thousands of sales transactions every day across multiple product categories, customer segments, and geographic regions. Raw transactional data stored in spreadsheets is nearly impossible for business managers to interpret quickly. Leaders constantly struggle to answer questions such as: Which product category is the most profitable? Which region is underperforming? Is the business growing month over month?

Without a centralized visual analytics tool, sales managers are forced to manually compile reports — a process that is time-consuming, error-prone, and outdated by the time it reaches decision-makers. This project addresses four core challenges:

- No quick, visual way to monitor KPIs such as Total Sales, Profit, Profit Margin, and Order Count
- Inability to track monthly revenue trends or identify seasonal patterns at a glance
- No geographic view of which US states and regions drive the most revenue
- Lack of an interactive tool allowing filter-based drill-down for non-technical business users

2. Solution / Features

A complete two-page interactive Sales Performance Dashboard was built in Power BI Desktop using the Kaggle US Superstore Sales dataset — 9,994 sales records across three product categories, four geographic regions, and nine distinct visual types. The solution transforms raw CSV data into an executive-ready analytics tool with zero coding required from end users.

Page 1 — Sales Overview

Delivers an executive-level snapshot of overall business performance. Four KPI cards instantly display Total Sales (725.46K), Total Profit (108.42K), Profit Margin (14.94%), and Order Count (3,203). A line chart tracks monthly sales trends throughout the year, revealing seasonal peaks and dips. A pie chart breaks down revenue by customer segment — Consumer (31.1%), Corporate (18.85%), and Home Office (50.05%). A scatter plot cross-references Total Sales vs. Total Profit by product category, and a horizontal bar chart ranks the three categories (Furniture, Technology, Office Supplies) by revenue. A Region slicer instantly filters all visuals for any combination of Central, East, South, and West.

Page 2 — Geographic & Category Deep Dive

Provides a state-level geographic breakdown and category profit analysis. A Bing bubble map of the United States plots total sales by individual state, immediately revealing top contributors such as California, New York, and Texas. A waterfall chart visualizes Total Profit by category — showing Technology as the primary profit driver, Office Supplies adding moderately, and Furniture contributing minimal margin. A matrix table at the bottom cross-references all four Regions against

nine Product Sub-Categories (Appliances, Art, Binders, Envelopes, Fasteners, Labels, Paper, Storage, Supplies) with row and grand total sales figures.

3. Dashboard Screenshots

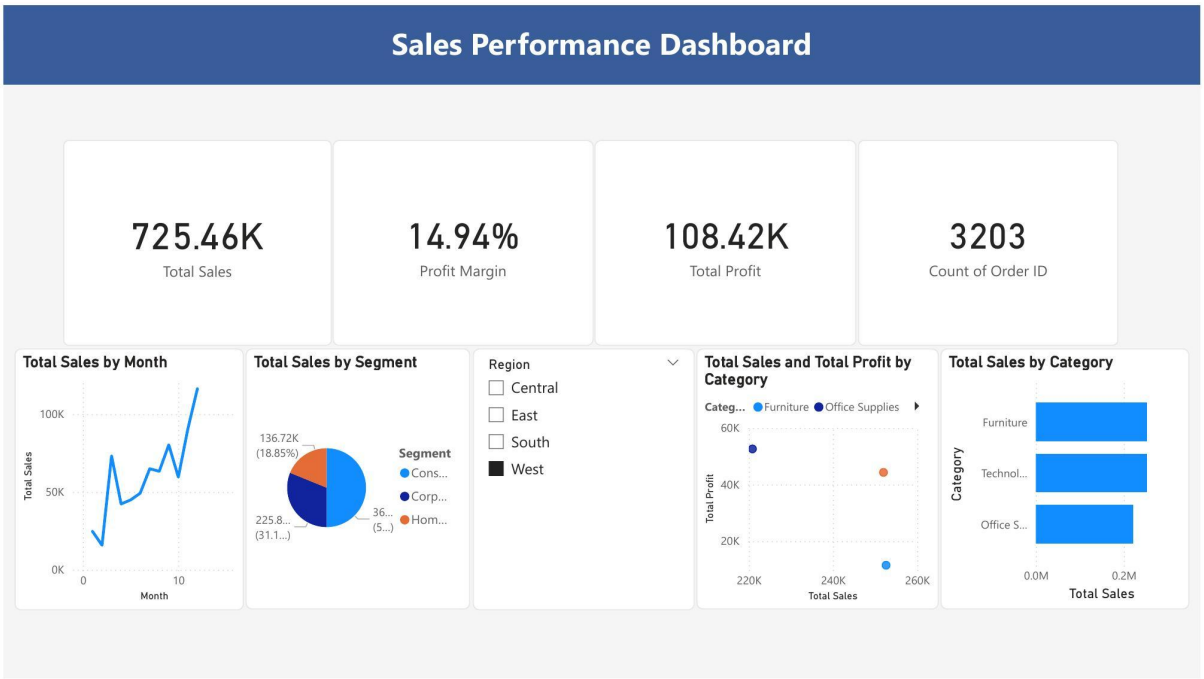

Figure 1: Page 1 — Sales Overview (KPI Cards, Monthly Trend, Segment Pie, Category Scatter & Bar)

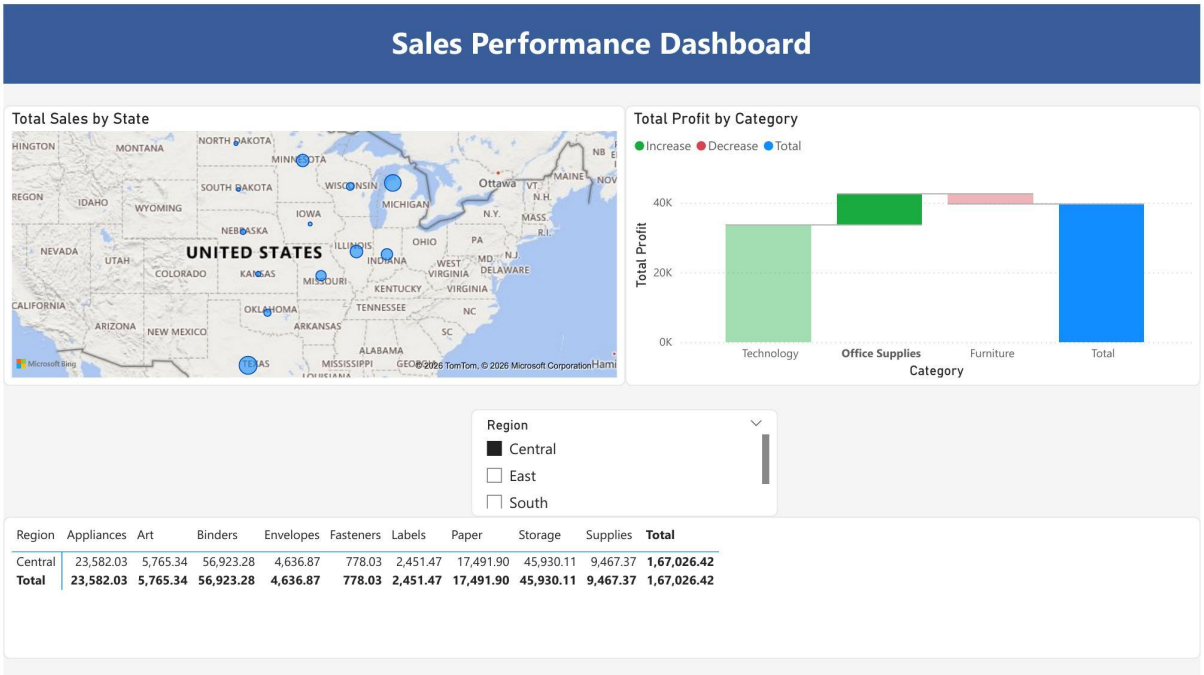

Figure 2: Page 2 — Geographic & Category Analysis (US Map, Waterfall Chart, Region × Sub-Category Matrix)

4. Tech Stack

Kaggle Dataset	US Superstore Sales — 9,994 rows covering Order ID, Date, Category, Sub-Category, Sales, Profit, Discount, Segment, State, and Region fields.
Python / Pandas	Data pre-processing: datetime formatting, duplicate removal, null handling, column standardization, and derived fields (Profit Margin, Days to Ship).
Power BI Desktop	Dashboard design, data modeling via Power Query, custom DAX Date Table using CALENDAR(), and nine interactive visuals across two pages.
DAX Measures	Total Sales, Total Profit, Profit Margin % (Profit / Sales), and Count of Orders — all powering the KPI cards and cross-filtered visuals.
GitHub	Hosts the Python notebook (.ipynb), cleaned CSV, Power BI .pbix file, and project report for evaluation access.

Dashboard Visuals Summary

KPI Cards	4 cards — Total Sales, Total Profit, Profit Margin, Order Count · Page 1
Line Chart	Total Sales by Month — monthly revenue trend across the year · Page 1
Pie Chart	Total Sales by Customer Segment (Consumer, Corporate, Home Office) · Page 1
Scatter Plot	Total Sales vs. Total Profit by Product Category · Page 1
Bar Chart	Total Sales by Product Category · Page 1
Region Slicer	Interactive filter — Central, East, South, West · Pages 1 & 2
Map Visual	Total Sales by US State — Bing geographic bubble map · Page 2
Waterfall Chart	Total Profit by Category (Increase / Decrease / Total) · Page 2
Matrix Table	Sales by Region × Sub-Category with grand totals · Page 2

5. Unique Points

✓ Waterfall Chart for Profit Decomposition: Uses a Waterfall chart instead of a plain bar chart to decompose profit into Increase (green), Decrease (red), and Total (blue) segments — immediately revealing that Furniture reduces overall profitability in a way a standard chart would not.

✓ Scatter Plot for Sales-Profit Relationship: Plots each product category at the intersection of Total Sales (X-axis) and Total Profit (Y-axis), exposing which categories are high-revenue but low-margin versus genuinely profitable — a two-dimensional view that ranked lists cannot provide.

✓ Region × Sub-Category Matrix with Totals: Cross-references all four regions against nine product sub-categories with row and column grand totals, giving a full 360-degree breakdown of where sales concentrate across the entire product catalogue.

✓ Geographic State-Level Map: Bubble map with size proportional to Total Sales reveals geographic concentration at a glance — coastal states like California and New York dominate — enabling targeted regional strategy decisions that tabular reports cannot surface quickly.

✓ Python-Cleaned Dataset: Unlike dashboards that import raw files directly, this project pre-processes data in Pandas — converting dates, removing duplicates, handling nulls, and engineering derived columns — ensuring analytical accuracy before any Power BI modeling.

✓ Cross-Page Synchronized Slicer: The Region slicer on both pages is synchronized so filtering by region on either page instantly updates all nine visuals, making the dashboard behave like a connected analytics application.

6. Future Improvements

- Customer RFM Analysis: A third dashboard page introducing RFM (Recency, Frequency, Monetary) scoring via DAX to automatically classify customers as Champions, Loyal, At-Risk, or Lost — enabling personalized sales outreach directly from the dashboard.
- Sales Forecasting with Python Integration: Embed a time-series forecasting model (Facebook Prophet or ARIMA) as a Power BI Python visual to predict the next 3–6 months of revenue based on historical monthly trends.
- Discount Impact Analysis: A dedicated page examining the relationship between discount rates and profit margins across categories and sub-categories — identifying which discount levels erode profitability versus drive effective volume growth.
- Live Data Connection: Replace the static CSV source with a direct Power BI connection to a SQL database or cloud source (Google Sheets or REST API) for automatic real-time dashboard refresh.
- Mobile-Optimized Layout: Design a responsive Phone Layout view in Power BI Desktop so that sales representatives can monitor KPIs on smartphones while in the field.

7. Conclusion

This project successfully demonstrates how Power BI can transform a raw retail CSV into an executive-ready, interactive analytics platform. By combining clean data engineering in Python with carefully designed Power BI visuals — including unconventional choices like the Waterfall chart and the Scatter plot — the dashboard delivers insights that static spreadsheets simply cannot match.

Key outcomes delivered: instant KPI visibility across Sales, Profit, Margin, and Orders; geographic identification of top-performing US states and regions; category-level profit decomposition revealing Furniture's margin drag; and a fully interactive, filter-driven experience requiring zero technical expertise from end users. The foundation built here is directly extensible toward real-time data, predictive analytics, and mobile access.